School of Information Technology and Engineering

Advanced PLC programming

Create Pump Control Unit (Experion PKS + UniSim Design)

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Introduction

The purpose of this laboratory work is to create and configure a pump control unit using Experion PKS Configuration Studio in connection with UniSim Design.

The main objective of the experiment is to establish communication between the simulation process and the controller, configure logical control, and analyze how the system responds to changes in operating conditions.

This work demonstrates the principles of process automation, logical control design, and system integration between UniSim and Experion PKS.

Objective

- To design a pump control unit using Experion PKS logic blocks.
- To simulate a process in UniSim Design with a pump and valves.
- To connect UniSim and Experion PKS through data exchange blocks.
- To verify the operation of the logical control (ON/OFF) of the pump.

Experimental Procedure

Step 1 – Process Creation in UniSim Design

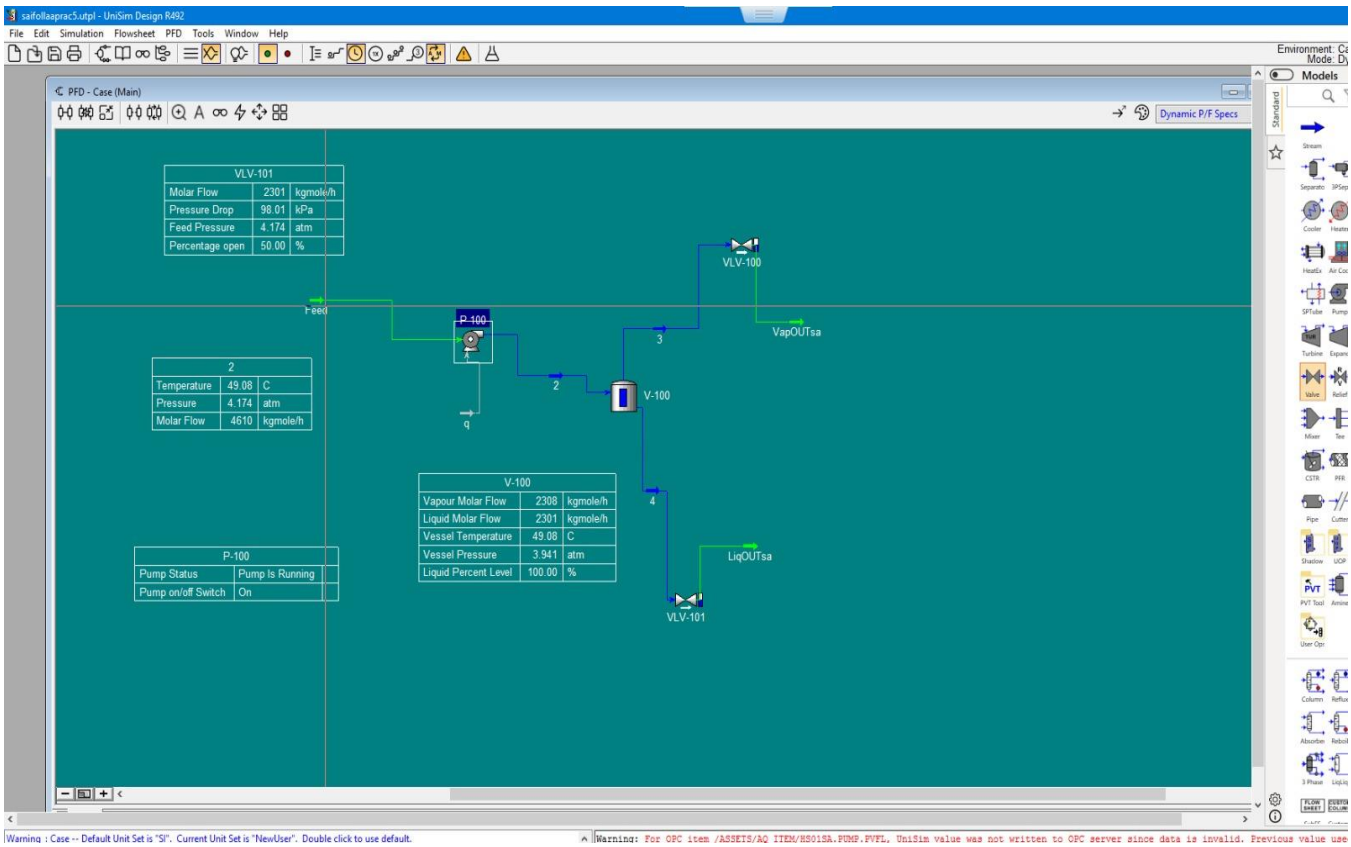
In UniSim Design, a simple process flow was developed consisting of:

- Pump (P-100) – the main controlled device;
- Valves (VLV-100, VLV-101) – used to regulate vapor and liquid flow;
- Vessel (V-100) – used for phase separation;
- Flow lines connecting all units.

The parameters of the process were monitored, including temperature (49.08°C), pressure (4.17 atm), and molar flow (4610 kgmole/h).

The pump's status (Running/Off) was displayed dynamically on the UniSim interface.

Figure 1 – Process Flow Diagram in UniSim Design showing pump, vessel, and valves.



Step 2 – Control Logic in Experion PKS

A control scheme was developed in Experion PKS Configuration Studio to manage the pump operation.

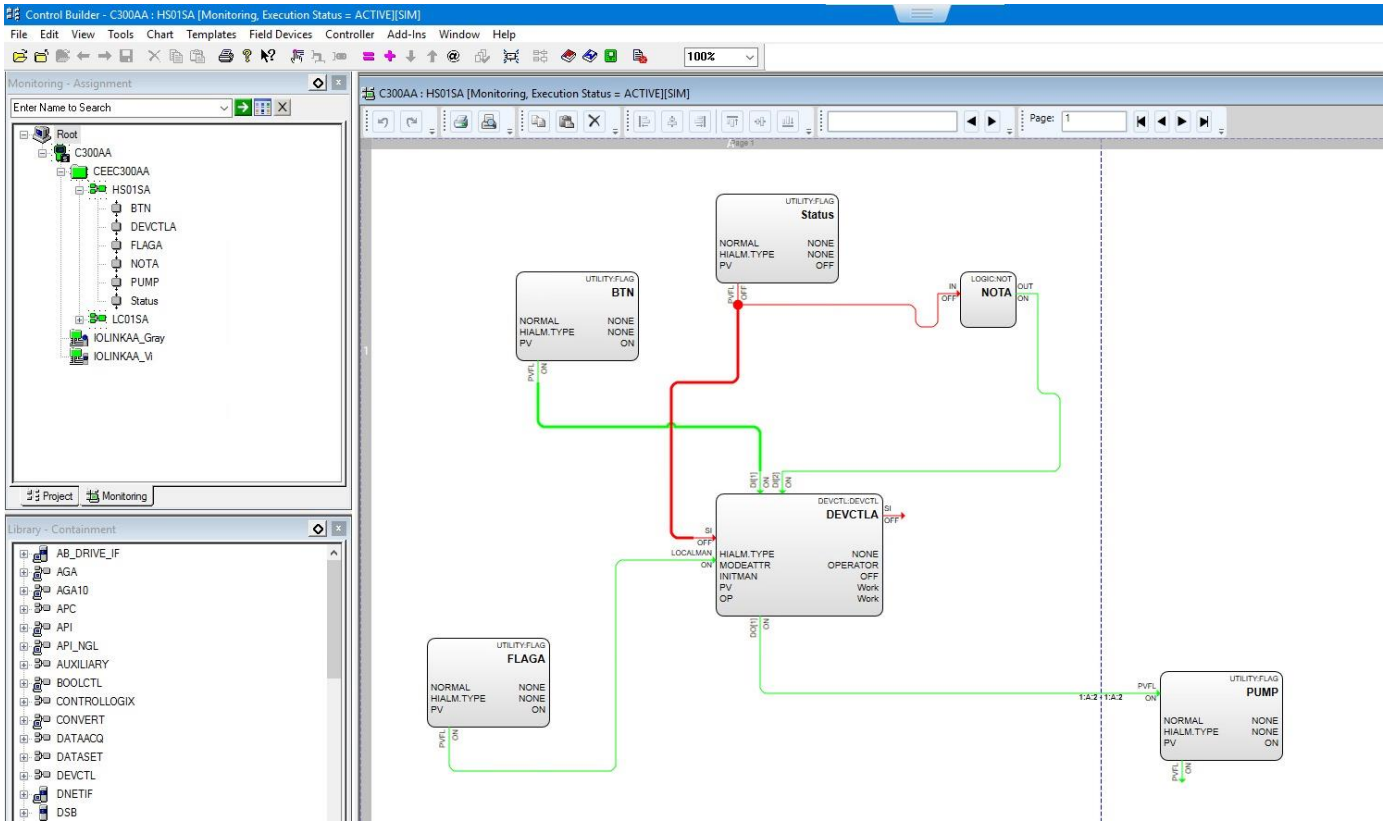
The following blocks were used:

- BTN (Button) – to manually start or stop the pump;
- STATUS – to indicate current system condition (ON/OFF);
- NOTA (Logic NOT) – to invert logic signal when needed;
- DEVCTLA – main device control block handling manual/auto control;
- FLAGA and PUMP – output flags showing active pump status.

The logical connections between blocks were configured so that pressing the button (BTN) changed the system state, activating or deactivating the pump.

Green lines indicate “active” signals, while red lines represent “inactive” signals.

Figure 2 – Experion PKS Control Logic Diagram for the Pump Control Unit.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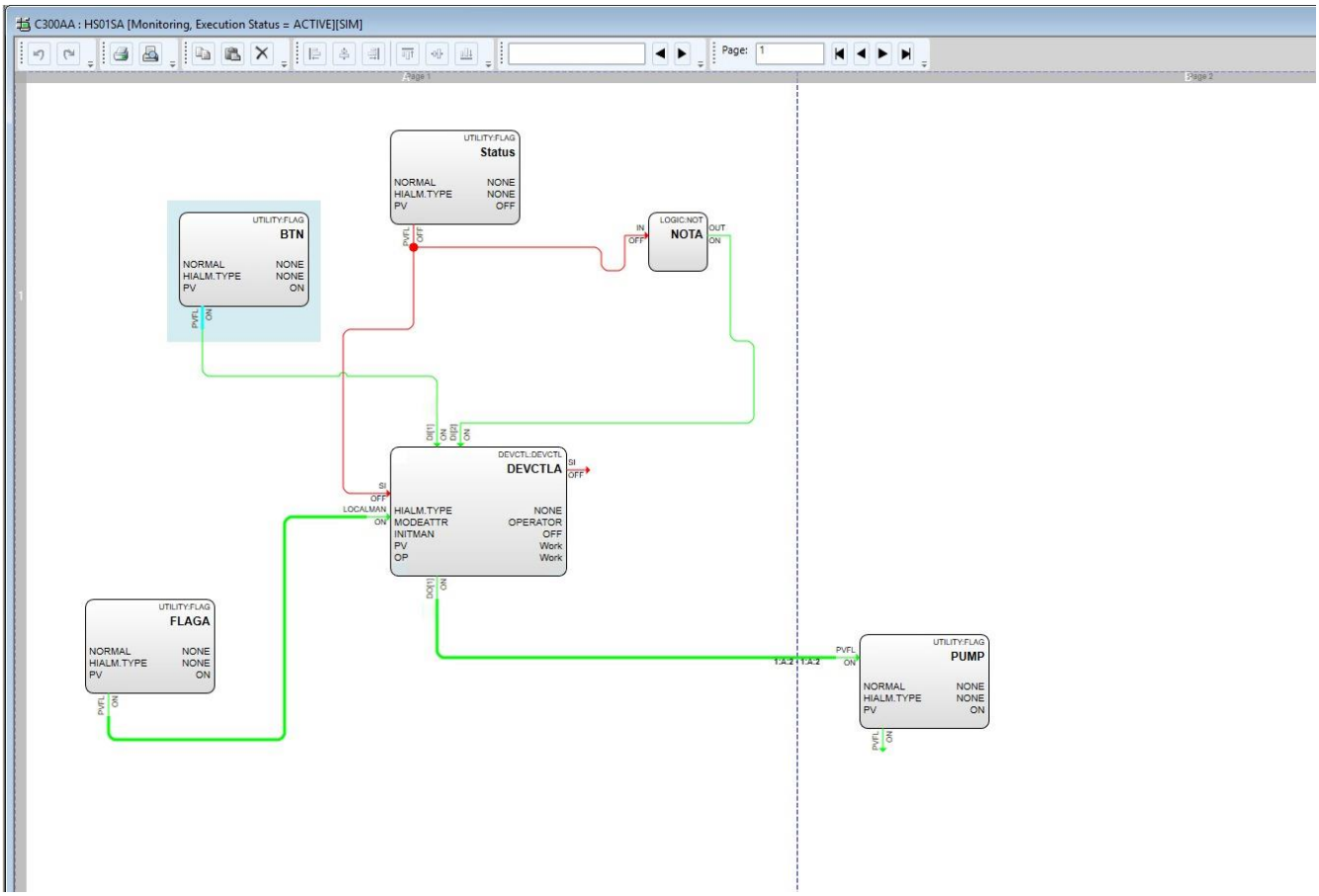
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FLAGA
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Step 3 – System Monitoring

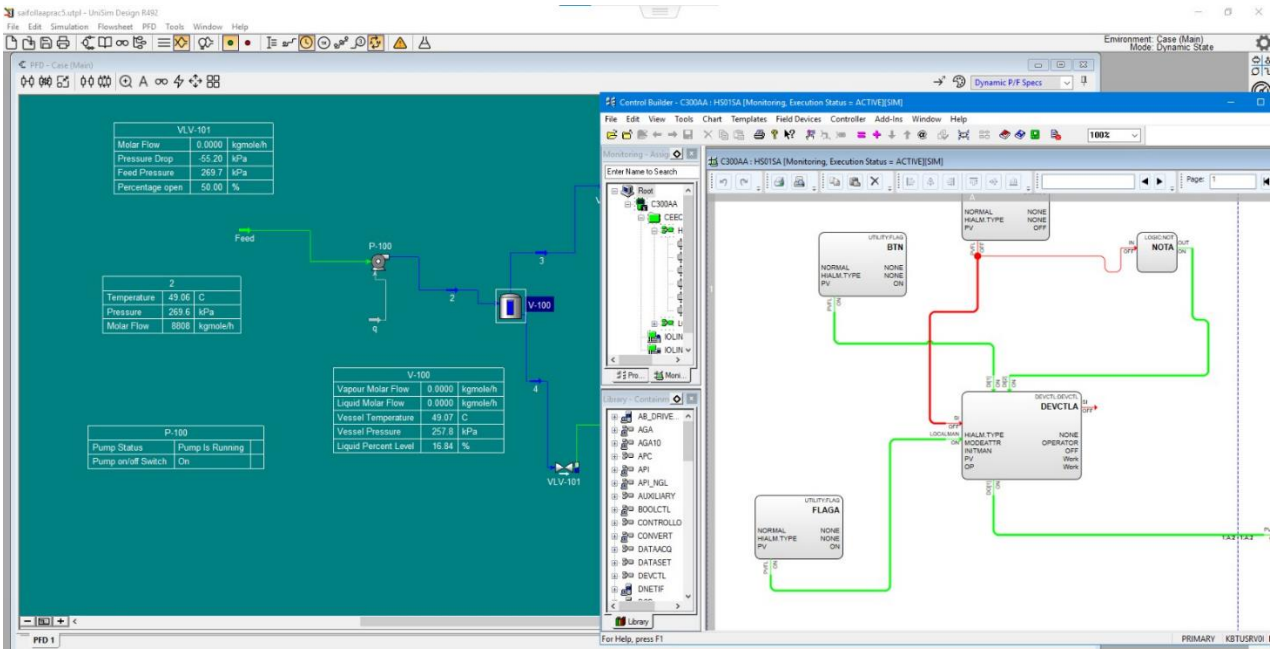
After configuration, the system was run in simulation mode (ACTIVE | SIM).

The logical status of each block was observed in real time.

When the BTN block was switched to ON, the PUMP flag became active, indicating that the pump was operating.

When the button was OFF, the NOTA block inverted the signal and stopped the pump.

The logical transitions and signal propagation confirmed that the control loop worked correctly.



Results and Discussion

• Communication between UniSim Design and Experion PKS was successfully established.

• Logical control of the pump was implemented using utility and device blocks.

• The simulation accurately reflected pump status changes based on the control signals.

• The system demonstrated stable operation and correct logical sequencing.

This setup can be extended to automatic control using PID loops or integrated into a larger process network.

Conclusion

In this laboratory work, a pump control unit was created and tested using Experion PKS and UniSim Design.

The logic scheme allowed manual control of the pump through button and flag signals.